For every n at least minus two, the type of n-types is an (n+1)-type

The Univalent Foundations Program

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Editorial note. The counted statement is the author thm environment at source lines 220-222 and its proof environment opens on line 223, the next line; the pairing is by adjacency and the retained proof range 223-234 is the whole environment.

Editorial note. The book label-number dump gives this label as 7.1.11 (env thm, chapter 7, section 1, eleventh counter value), so the counters chapter 7, section 1 and thm 10 (zero-based) are set so that the item prints with the book number.

Editorial note. The proof invokes an observation stated immediately above the counted theorem (the displayed equivalence of the identity type in the universe of n-types with the equivalence type); that display is retained as context so that the step is visible in the card, and the label it cites is declared as an external reference.

Editorial note. The first line of the proof carries a commented-out proof heading in the source; it is retained verbatim and has no effect on the rendering.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the type of n-types is an (n+1)-type (author label thm:hleveln-of-hlevelSn, book Theorem 7.1.11).

Allowed standard prerequisites (editorial).

- The universe of n-types. The type of n-types is the dependent pair of a type and a proof that it is an n-type, and the identity type of such a pair is equivalent to the type of equivalences between the underlying types because being an n-type is a mere proposition.. It is the type whose (n+1)-type property the counted theorem asserts, and its identity type is the first object analysed in the proof.
- Embeddings and n-type-ness of function types. The projection from equivalences between X and X-prime to functions from X to X-prime is an embedding, and function types into an n-type are n-types.. The proof reduces the goal to the function type and then to the assumption that the target is an n-type.

- Contractible types and the unit type. Any two contractible types are equivalent to the unit type and hence to each other, so the type of equivalences between them is inhabited.. It is the only content of the separate case n equal to minus two.

Source reference. Original theorem 7.1.6: sources/hott-hlevels/pinned-original-source.tex, source lines 102–102. Book Theorem 7.1.6, monotonicity of n-typeness along embeddings, invoked in the retained proof to move from the equivalence type to the function type. Navigation pointer only: the lemma is counted by its own card of this lane.

Source reference. Original theorem 7.1.8: sources/hott-hlevels/pinned-original-source.tex, source lines 467–467. Book Theorem 7.1.8, on path types over a subset, cited by the retained context display to justify replacing the identity type in the universe of n-types by the equivalence type. Navigation pointer only: it is counted by its own card of this lane.

Note that just as for Prop and Set, because is-n-type(X) is a mere proposition, by original theorem 7.1.8 for any $(X, p), (X', p') : n\text{-Type}$ we have

$$\begin{aligned}\Big((X,p) =_{n\text{-Type}} (X',p')\Big) &\simeq (X =_{\mathcal{U}} X') \\ &\simeq (X \simeq X').\end{aligned}$$

Theorem 7.1.11. *For any $n \geq -2$, the type n-Type is an $(n+1)$-type.*

Proof. Let $(X, p), (X', p') : n\text{-Type}$; we need to show that $(X, p) = (X', p')$ is an n-type. By the above observation, this type is equivalent to $X \simeq X'$. Next, we observe that the projection

$$(X \simeq X') \to (X \to X').$$

is an embedding, so that if $n \geq -1$, then by original theorem 7.1.6 it suffices to show that $X \to X'$ is an n-type. But since n-types are preserved under the arrow type, this reduces to an assumption that X' is an n-type.

In the case $n = -2$, this argument shows that $X \simeq X'$ is a (-1)-type — but it is also inhabited, since any two contractible types are equivalent to $\mathbf{1}$, and hence to each other. Thus, $X \simeq X'$ is also a (-2)-type. □